

Cyber Defense (420-D03-AB)

Final Project and Course Procedure Manual

Program	Cybersecurity AEC (Attestation d'études collégiales) — LEA.D8
Competency	EZ03 — To apply access control measures
Author and owner	Cyber.SoHo
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Presented at	The first held class of the session — Tuesday, September 1, 2026
Governing document	Legal Notice, Disclaimer, and Acceptable Use Policy — Cyber Defense (420-D03-AB), Version 1.0, effective August 29, 2026 — reproduced in full in Annex A

Session note. The class of Saturday, August 29, 2026 (Class 1) was cancelled. Its content is delivered together with today's class. Every deliverable, due date, and grade weight in this document is unchanged.

This document contains, in two sections, the complete and final requirements of the two graded deliverables that close the course:

- **Section 1 — Final Project (15% of the course grade)** — due Class 21, Tuesday, October 20, 2026.
- **Section 2 — Course Procedure Manual (10% of the course grade)** — due Class 21, Tuesday, October 20, 2026.

Nothing will be added to these requirements later. The grading checklists are printed inside this document (§ 1.7 and § 2.9); the identical checklists are re-distributed at Class 18, unchanged, so there are no hidden expectations.

0. Read This First — The Legal Framework That Applies to Everything Below

0.1 Educational purpose only. All lectures, labs, projects, quizzes, exams, documents, videos, and social-media publications of the course Cyber Defense (420-D03-AB) — together, "the Material", which includes this document and both deliverables it describes — are created for one purpose: teaching defensive cybersecurity to adult learners of the Cybersecurity AEC (Attestation d'études collégiales) program. The Material shows how to protect systems, not how to attack the systems of others. Nothing in the Material is an invitation, an instruction, or an authorization to access, test, or damage any real system, network, account, or data.

0.2 Plain-language summary of your obligations. This framework protects everyone: the students, the viewers, and the author. In short: all the material of this course is provided for education only. You may practice the activities only inside your own, isolated laboratory environment, or on systems for which you hold explicit written permission. You are fully and solely responsible for your own actions, your equipment, your accounts, your costs, and your time. By reading, watching, downloading, or using any part of the material — including this document — you accept the entire Legal Notice, Disclaimer, and Acceptable Use Policy (Annex A).

0.3 Acknowledgment and acceptance — the session signature. Enrolled students also confirm their acceptance electronically: at the beginning of **each** session, every student signs the separate "Acknowledgment and Agreement" page, which creates the student's personal electronic signature record for that session. The signing page (`Acknowledgment_and_Agreement_eSign.html`) and its signature log (`signatures.json`) are provided separately on the course platform.

0.4 The one rule that can never be broken. You practice only inside your personal, isolated laboratory environment (as defined in LAB_01), or on systems for which you hold explicit written permission obtained **before** you start. Unauthorized computer access or use is a criminal offence — in Canada, the Criminal Code contains, among other provisions, section 342.1 (unauthorized use of a computer), section 342.2 (possession of a device to obtain unauthorized use of a computer), and section 430(1.1) (mischief in relation to computer data), and penalties can include imprisonment. If you are located outside Canada, the laws of your own country, province, or state apply to you, and you must know and obey them. The policies and codes of conduct of your college apply in addition to the law. The instructor never asks for, and never authorizes, any illegal act. **If an activity seems to touch a system you do not own, stop and ask the instructor before doing anything.**

0.5 Where the rules live in this document. The complete Legal Notice (16 sections) is reproduced word for word in Annex A. Its operational rules are also embedded at the exact places where you will apply them: laboratory and acceptable-use rules in § 1.4 and § 2.4; the AI (Artificial Intelligence) disclosure rule in § 1.8 and § 2.6; intellectual property, warranty, liability, links, social media, privacy, and governing law in § 1.10 and on your manual's cover page (§ 2.3).

0.6 Where these two deliverables sit in the evaluation plan.

Evaluation	Weight	When
Quizzes Q_01 to Q_04	$4 \times 2.5\% = 10\%$	Classes 2, 9, 14, 18
Hands-on labs LAB_01 to LAB_06	$6 \times 5\% = 30\%$	Throughout the course
Mid-Term Exam	10%	Class 10
Final Project (Section 1 of this document)	15%	Due Class 21
Course Procedure Manual (Section 2 of this document)	10%	Due Class 21
Final Exam	25%	Class 23
Total	100%	

SECTION 1 — FINAL PROJECT (15%)

Due: Class 21 — Tuesday, October 20, 2026, at the start of class.

1.1 Goal — One Small, Fully Defended Company

You deliver **one small, fully defended company IT (Information Technology) infrastructure**, built and validated inside your own isolated laboratory. Nothing in the project is new: it is the direct assembly of the six hands-on labs (LAB_01 to LAB_06) completed during the course, plus **one** short resilience assessment — the only new element. If you keep up with the labs and document them the week they are done, the Final Project is essentially finished by Class 17; Classes 18 to 21 are reserved for assembly, the resilience assessment, and clean documentation.

The project demonstrates the course competency **EZ03 — To apply access control measures**: every chapter shows a layer of defense-in-depth (layering several simple protections instead of relying on one) that controls who and what may access the company's systems, networks, and data.

1.2 The Fictional Company — The Same Company in Every Lab and in the Project

All labs and the Final Project defend the same small fictional company. Its official profile follows. You may rename the company and adjust minor details for creativity, provided the role table and the minimum system list stay equivalent, so that every lab instruction remains applicable.

Company profile — NorthMaple Accounting Inc. (fictional). A small accounting firm of about ten people, in a single office, holding sensitive client financial records. It has one internet connection, a public company website, an internal file server, employee workstations, and wireless access for staff and visitors. One or two employees work remotely and must reach internal data securely.

Role table (used in LAB_02 and throughout):

Role	Count	What this role needs to access
Administrator (IT)	1	Full, controlled administrative access on all systems (via admin group / sudo)
Accountant	3	The client-records folder (read/write); no administrative rights
Technician	2	The technical-documentation folder; limited administrative rights on workstations only
Receptionist	1	The front-desk shared folder only

Role	Count	What this role needs to access
Remote worker	1–2	The same access as their role, reachable only through the VPN (Virtual Private Network)

Minimum system list (all as virtual machines or free-tier cloud instances):

1. One network firewall/router appliance — pfSense or OPNsense — at the perimeter (LAB_05).
2. One public web server (the company website) placed in the DMZ (Demilitarized Zone).
3. One internal file server (Windows or Linux) in the server zone.
4. One Windows workstation and one Linux workstation in the internal user zone.
5. One wireless access point (may be simulated/documented) with a corporate SSID (Service Set Identifier) in the user zone and a guest SSID in its own guest zone (added at Class 13).
6. One IDS (Intrusion Detection System) sensor — Snort or Suricata (LAB_04).
7. One "remote worker" client machine that connects only through the VPN (LAB_06).

1.3 Assembly Map — How the Six Labs Become the Project

Each completed lab becomes one chapter of the Final Project, and its documentation becomes the matching chapter of the Course Procedure Manual (Section 2). Documenting each lab **the same week it is done** means no extra work at the end of the course.

Project chapter	Built from	Built during	Lab weight	Chapter deliverable
1. Defensible architecture	LAB_01	Classes 1–2	5%	Zoned network diagram + zoning table
2. Access control	LAB_02	Classes 3–4	5%	Users, groups, least-privilege permissions + validation
3. Hardened hosts	LAB_03	Classes 5–6	5%	Hardening checklist results + host firewall rules
4. Active defense	LAB_04	Classes 7–8	5%	Running IDS, rule set, captured alerts
5. Network security	LAB_05	Classes 11–12	5%	Network firewall enforcing the zoning + device registry
6. Secure communication	LAB_06	Classes 16–17	5%	Working VPN + encryption of stored data + validation
7. Resilience assessment	New (template)	Classes 18–19	part of 15%	Completed resilience checklist

1.4 Working Rules That Apply to Every Part of the Project

These rules come from section 3 (Acceptable Use and Prohibited Actions), section 4 (Legal Warning), and section 11 (Ethics and Responsible Conduct) of the Legal Notice. They apply to every lab, every test, and every screenshot in your project.

You may:

- Practice **only** inside your personal, isolated laboratory environment, as defined in LAB_01: a local virtualization tool (for example VirtualBox) or your own personal free-tier cloud account on AWS (Amazon Web Services) or Microsoft Azure.
- Practice on any other system **only** if you hold explicit **written** permission from the owner of that system, obtained before you start.
- Use snapshots and isolated virtual networks so that a mistake stays inside your laboratory.

You must never:

- Scan, probe, attack, intercept, monitor, or access any system, network, wireless network, or account that is not yours — including those of the college, an employer, a business, a public website, or a neighbor.
- Create, install, or release malicious software outside your isolated laboratory.
- Attempt password, credential, or interception attacks against any real account or any real communication.
- Use the Material, or any tool presented in it, to break any law or any policy that applies to you.

Cloud responsibilities. If you use a cloud account, you must respect the acceptable-use policies of the provider (AWS or Microsoft Azure). Free tiers have limits; monitor them. All cloud costs, overage charges, and account consequences are your responsibility.

Ethics and responsible conduct. The course teaches defensive ("white-hat") cybersecurity. You agree to use your skills only to protect, never to harm, and to respect the privacy of others. If you accidentally discover a real vulnerability, report it responsibly to the owner of the system and to the instructor, instead of exploiting it or publishing it. All test traffic in this project (for example, the IDS validation traffic of Chapter 4 and the firewall tests of Chapters 3 and 5) is generated **inside your own laboratory only**, against your own virtual machines only.

1.5 Chapter-by-Chapter Requirements

Each chapter below states what must exist and work in your laboratory, and what evidence proves it. The written procedure for each chapter lives in the Course Procedure Manual (Section 2, § 2.5); the project chapter carries the working configuration and its proof.

1.5.1 Chapter 1 — Defensible Architecture (from LAB_01)

What must exist:

1. Your personal, isolated laboratory environment, set up and documented (VirtualBox, or a personal free-tier AWS or Azure account), with snapshots and an isolated virtual network.
2. The complete network diagram of NorthMaple Accounting Inc., showing every system of the minimum list, all perimeter elements and entry points, and the zones: Internet, DMZ, internal user zone, server zone — plus the guest Wi-Fi zone added at Class 13.
3. The **zoning table**: every company system placed in exactly one zone, and every allowed traffic flow between zones listed with its source zone, destination zone, service/protocol, and a justification of one or two plain sentences.

Evidence required: the finished diagram; the zoning table; environment-setup documentation (first pages of the manual).

Validation: no system without a zone; no inter-zone flow without a written justification; trust levels between zones stated.

1.5.2 Chapter 2 — Access Control (from LAB_02)

What must exist:

1. Users and groups created on one Windows machine and one Linux machine, exactly per the role table of § 1.2 (administrator, accountant, technician, receptionist, remote worker).
2. Least-privilege and need-to-know group memberships: each role reaches only what it needs.
3. Folder permissions per role: NTFS (New Technology File System) permissions and sharing on Windows; file ownership, permission bits, and a controlled sudo policy on Linux.

Evidence required: the completed permission tables; validation screenshots of simple test logins per role; the completed validation checklist.

Validation: each role can open its own data and is refused everywhere else — proven by the test-login screenshots.

1.5.3 Chapter 3 — Hardened Hosts and Host Firewalls (from LAB_03)

What must exist:

1. The provided hardening checklist (about ten items per system) applied to your Windows and Linux machines: accounts, services, updates, and security policies on Windows; services, SSH (Secure Shell) settings, and updates on Linux — with configurations checked against free baselines such as the CIS (Center for Internet Security) Benchmarks.
2. A **before-and-after** record proving each hardening step was applied.
3. Host-based firewall rules matched to each machine's role (workstation versus server): Windows Defender Firewall on Windows; UFW (Uncomplicated Firewall) or firewallld on Linux — a small set of inbound rules per host.

Evidence required: hardening checklist results with before/after proof; the firewall rule tables; connectivity-test evidence.

Validation: each rule proven with a simple allowed/blocked connectivity test, results recorded.

1.5.4 Chapter 4 — Active Defense: Intrusion Detection (from LAB_04)

What must exist:

1. One free, open-source IDS — Snort or Suricata, your choice — installed on a lab machine, running, and confirmed to see traffic.
2. The small provided rule set enabled; you may add simple rules of your own.
3. Harmless test traffic generated **inside your own lab only**, and the resulting alerts captured.

Evidence required: installation notes; the rule set used; alert screenshots.

Validation: at least one captured alert corresponding to your own safe, contained test traffic, demonstrating that the detection system actually detects.

1.5.5 Chapter 5 — Network Security: Firewall and Device Registry (from LAB_05)

What must exist:

1. A free, open-source firewall/router appliance — pfSense or OPNsense, your choice — deployed as a virtual machine in front of your company network.
2. Its management access secured: strong credentials, HTTPS (Hypertext Transfer Protocol Secure) management, SSH only where needed; default credentials removed; firmware/version recorded.
3. Firewall rules that **enforce the Chapter 1 zoning table**, translating every allowed and denied inter-zone flow into a real rule.

4. The **company device registry**: every network device and appliance with its role, model or virtual machine, software version, management access method, and update status — the start of a configuration-management plan.

Evidence required: the firewall rule set; the device registry; allowed/blocked validation evidence for every rule.

Validation: each rule tested with a simple allowed/blocked connection test that matches the zoning table.

1.5.6 Chapter 6 — Secure Communication: VPN and Encryption (from LAB_06)

What must exist:

1. A working remote-access VPN into the company network — WireGuard recommended for simplicity; OpenVPN or the built-in VPN features of pfSense/OPNsense accepted — with keys, users, and allowed networks configured.
2. A "remote worker" client connected through the VPN and verified.
3. Encryption of stored data: one folder or volume of company data encrypted with a free tool.
4. HTTPS enabled with a self-signed certificate on one internal service, showing how PKI (Public Key Infrastructure) concepts — certificates and CAs (Certificate Authorities) — apply in practice.

Evidence required: the VPN configuration; the encryption steps; the certificate/HTTPS configuration; validation evidence.

Validation: VPN traffic is shown to be encrypted; the internal service answers **only** over HTTPS; the encrypted folder/volume is proven inaccessible without its key.

1.5.7 Chapter 7 — Resilience Assessment (the only new element)

A short, template-based checklist evaluating how your defended network withstands common attack scenarios, with **one plain-language sentence of justification per item**, naming the defense layer(s) from Chapters 1 to 6 that respond. This is how the course assesses and documents network resilience: the ability to keep the business running during and after an attack. It is completed during Classes 18 and 19.

Official template (fill every row):

#	Attack scenario (happens inside the fictional scenario only)	Defense layer(s) that respond (Ch. 1–6)	Verdict: Withstands / Partly / Does not	One-sentence justification
1	A phishing email plants malware on one user workstation			
2	Repeated password guessing against the remote access			
3	The public web server in the DMZ is compromised			
4	A worm tries to spread across the internal network			
5	A company laptop with client data is lost or stolen			
6	A remote worker's traffic is intercepted on public Wi-Fi			
7	A DoS (Denial-of-Service) attempt targets the public service			
8	An accountant tries to open the technicians' and administrator's data			
9	An attacker tries default credentials on the network devices			
10	Ransomware attempts to encrypt the file server			

Reminder (No Professional Advice — Legal Notice, section 7): this assessment is an educational exercise on a fictional company. It is not professional security consulting; an organization that needs to protect a real infrastructure must consult qualified professionals.

1.6 What You Submit (the Package)

One archive, submitted on the course platform at Class 21, containing:

1. **The Course Procedure Manual** (Section 2 of this document) — one file, PDF (Portable Document Format) or Word.
2. **The evidence set** — the screenshots and configuration exports gathered throughout the course, organized in one folder per chapter (CH1_architecture/ to CH7_resilience/).
3. **The completed resilience assessment** (the § 1.5.7 template), either as its own file or as the manual's resilience chapter.

4. **The AI disclosures** — inside each chapter of the manual (§ 2.6); nothing separate is needed if every chapter carries its table or its "no AI used" statement.

Naming convention: 420-D03-AB_C21_FinalProject_LASTNAME_Firstname.zip; the manual inside is named 420-D03-AB_C21_Manual_LASTNAME_Firstname.pdf (or .docx).

The working lab environment itself stays on your machine or cloud account; the evidence set is what proves it works. Keep your environment intact until final grades are released, in case a verification is requested.

1.7 Grading Checklist — Final Project (15 points = 15%)

This is the complete grading checklist. The identical checklist is re-distributed at Class 18, unchanged. Within each chapter: **1.0 point** for "it exists and works" (the evidence), **0.5 point** for "it is validated" (the tests), **0.5 point** for "it is justified" (the plain-language reasoning).

Item	Points
Chapter 1 — Defensible architecture: diagram, zoning table, environment	2.0
Chapter 2 — Access control: roles, least privilege, permissions, test logins	2.0
Chapter 3 — Hardened hosts: checklist with before/after, host firewall rules, tests	2.0
Chapter 4 — Active defense: running IDS, rule set, captured alerts	2.0
Chapter 5 — Network security: firewall enforcing the zoning, device registry, tests	2.0
Chapter 6 — Secure communication: VPN, data-at-rest encryption, HTTPS, validation	2.0
Chapter 7 — Resilience assessment: all ten rows, one sentence each	2.0
Assembly and coherence: all six components run together; gaps listed at Class 18 are fixed; naming and package complete	1.0
Total	15.0

A missing AI disclosure where an AI tool was used is an academic-integrity issue (§ 1.8), handled under the college's policies — it is not a simple point deduction.

1.8 AI and Creativity Rules (Plagiarism Avoidance)

Creativity is welcomed, within the academic-integrity rule of the Legal Notice (section 11):

1. **You may use AI tools at any stage** — provided each chapter of your submission states which LLM (Large Language Model) was used and the **exact prompt(s)** given to it. This is the course's plagiarism-avoidance requirement, and it applies to every chapter of the project and the manual. The disclosure format is given in § 2.6.
2. **You may substitute equivalent open-source tools** for the suggested ones (for example, another free IDS in place of Snort/Suricata), provided the chapter's objective and validation are still met and the substitution is stated.

3. **You may build your own small helper scripts or tools** where applicable — they are welcome and expected, and they belong to you.
4. **You verify before you apply.** Anything an AI tool produces is checked by you inside your isolated laboratory before it touches your configuration; you remain fully and solely responsible for your own actions (Legal Notice, sections 1 and 5).

1.9 Timeline and Milestones

Class — date	Milestone for the project and manual
C1–C2 — Aug 29 (cancelled) + Sep 1	LAB_01 done; manual started (environment + Chapter 1)
C3–C4 — Sep 2, Sep 8	LAB_02 done; Chapter 2 written
C5–C6 — Sep 9, Sep 12	LAB_03 done; Chapter 3 written
C7–C8 — Sep 15, Sep 16	LAB_04 done; Chapter 4 written
C9 — Sep 19	Catch-up: Chapters 1–4 documented and up to date
C10 — Sep 22	Mid-Term Exam (no project work due)
C11–C12 — Sep 23, Sep 26	LAB_05 done; Chapter 5 written
C13 — Sep 29	Guest Wi-Fi zone added to the design; firewall rules reviewed
C14–C15 — Sep 30, Oct 6	Cryptography preparation; LAB_06 environment verified
C16–C17 — Oct 7, Oct 10	LAB_06 done; Chapter 6 written
C18 — Oct 13	Project assembly starts; grading checklist re-issued; gap list made
C19 — Oct 14	Gaps fixed; resilience assessment (Chapter 7) completed
C20 — Oct 17	Everything finalized; voluntary dry-run against the checklist
C21 — Oct 20	Final Project and Course Procedure Manual DUE; submission confirmed in class

1.10 Legal Terms That Apply to the Project and to Your Work

These clauses of the Legal Notice apply directly to the two deliverables; their full text is in Annex A.

Intellectual property (section 8). © 2026 Cyber.SoHo — Cyber.SoHo is the author and exclusive owner of the Material and of its structure, texts, diagrams, labs, templates, and supporting files. As an enrolled student you hold a limited, personal, non-transferable, non-exclusive license to use the Material for your own study; it does not allow redistribution, re-publication, resale, public posting, or commercial use, in whole or in part, without prior written permission. **The work you create in the course — your labs, your project, your manual — belongs to you;** the instructions and templates it is based on remain the property of Cyber.SoHo.

Copyright independence and no affiliation (section 9). Cybersecurity uses a shared professional vocabulary ("defense-in-depth", "least privilege", "network zoning"); its use implies no link with any other author or training body. Cyber.SoHo has no affiliation with, sponsorship by, or endorsement from any company, organization, product, service, or platform named in the Material, including, without limitation: Oracle (VirtualBox), Amazon (AWS), Microsoft (Azure, Windows), Linux distributions and their publishers, Cisco (Snort), the OISF (Open Information Security Foundation) (Suricata), Netgate (pfSense), Deciso (OPNsense), WireGuard, OpenVPN, the OpenSSL project, the CIS, Google (YouTube), and TikTok. All product names, logos, and brands are trademarks of their respective owners, used for identification and education only.

Assumption of risk and limitation of liability (section 5). You use the Material at your own risk, on your own or rented equipment, and on your own or class time. To the maximum extent permitted by law, Cyber.SoHo and the instructor — in every role: teacher, author, instructor, researcher, and publisher on social platforms — are not responsible for any direct or indirect damage, data loss, hardware or software failure, service or account suspension, financial cost, or legal consequence resulting from the use or the misuse of the Material, by you or by anyone else.

No warranty (section 6). The Material is provided "as is" and "as available", without any warranty of any kind. Tools, interfaces, prices, links, and best practices change over time; accuracy, completeness, and fitness for a particular purpose are not guaranteed.

No professional advice (section 7). The Material is educational — it is not legal advice and not professional security consulting. For any question about your own legal situation, consult a lawyer licensed in your jurisdiction.

External links (section 10). Links in the Material are provided for convenience only and are not endorsements; Cyber.SoHo does not control external websites. Before opening any external link, you are invited to verify it with a free online scanner, for example the VirusTotal URL (Uniform Resource Locator) scanner: <https://www.virustotal.com/gui/home/url>.

Social-media notice (section 12). Clips, posts, shorts, and videos of the course published on public platforms (for example YouTube or TikTok) are educational demonstrations performed inside an isolated laboratory — do not reproduce them outside such a laboratory. If you record your own lab for a class presentation, the same rule applies to your recording. Third-party comments, re-uploads, edits, reactions, and translations are not verified and not endorsed by Cyber.SoHo.

Indemnification (section 13). By using the Material, you agree to indemnify and hold harmless Cyber.SoHo and the instructor from any claim, demand, damage, loss, or expense (including reasonable legal fees) resulting from your own actions or omissions, from your violation of the Legal Notice, or from your violation of any law or of any right of a third party.

Privacy (section 14). The Material itself collects no personal data. When you sign the "Acknowledgment and Agreement" page, your name, the session number, the date, and a verification code are recorded in the course signature log; the drawn signature image stays only inside the record file that you download and submit, and it is not published. If the signature log is

hosted in a public repository, the log is publicly visible; private hosting is recommended. Keep personal data out of your own deliverables too — see the documentation standard, § 2.7.

Governing law, severability, and updates (section 15). The Legal Notice is governed by the laws applicable in the Province of Quebec, Canada; if a court finds one clause invalid, the others remain in force; Cyber.SoHo may update the document at any time, the current version replacing all previous versions.

SECTION 2 — COURSE PROCEDURE MANUAL (10%)

Due: Class 21 — Tuesday, October 20, 2026, together with the Final Project package.

2.1 Purpose — The Manual a Real Technician Would Leave Behind

The Course Procedure Manual is the professional documentation of the six labs: the kind of manual a real technician would leave for a colleague, so that **every procedure can be reproduced** exactly, by someone who was not there. If a classmate could rebuild your defended company by following your manual alone, inside their own isolated laboratory, the manual has done its job.

2.2 The Golden Rule — Write It During the Course

The manual is written **during** the course, one chapter per lab, in the supervised student work time already built into every class (weekday classes after the 7:30 pm break; Saturday classes after the 10:30 am break) — never at the end. Combined with the course ponderation of 3-3-3 (3 hours lecture, 3 hours practical work, 3 hours homework per week), a student who keeps up with the labs finishes both deliverables inside the scheduled time, without overload. The writing schedule is the milestone table of § 1.9: each chapter is written the same week its lab is completed, and Class 9 is a built-in catch-up point for Chapters 1 to 4.

2.3 Mandatory Front Matter (the Cover Pages)

Your manual opens with, in this order:

1. **Cover page:** course name and number (Cyber Defense, 420-D03-AB), program (Cybersecurity AEC — LEA.D8), competency (EZ03 — To apply access control measures), the manual's title, your full name (and student number if you have one), the session (Fall 2026), and the submission date (Class 21 — October 20, 2026).
2. **The mandatory notice box** — copy this text exactly:

This manual documents educational laboratory work performed entirely inside my own, isolated laboratory environment, in accordance with the Legal Notice, Disclaimer, and Acceptable Use Policy — Cyber Defense (420-D03-AB), Version 1.0, which I have read, accepted, and signed at each session. Nothing in this manual is an invitation, an instruction, or an authorization to access, test, or damage any real system, network, account, or data. It is an educational document, not professional security consulting.

3. **The copyright line** — copy and complete:

Documented procedures © 2026 [Your full name]. Course instructions and templates © 2026 Cyber.SoHo, used under the course's limited, personal, non-transferable student license — no redistribution, re-publication, resale, public posting, or commercial use without prior written permission.

4. **Short description of the fictional company** (from § 1.2): who it is, what it protects, its role table, and its minimum system list — plus any renaming or minor adaptation you made.
5. **Table of contents.**

2.4 The Lab Environment Chapter

Started at LAB_01 and kept up to date, this chapter describes the environment every procedure runs in:

1. Your platform choice: local virtualization (for example VirtualBox) **or** your personal free-tier cloud account (AWS or Microsoft Azure) — and why.
2. The virtual machine list: name, operating system, role in the company, zone, and resources.
3. The isolation measures: isolated virtual networks and snapshots, so that a mistake stays inside the laboratory (Legal Notice, section 3).
4. If cloud is used: the free-tier limits you monitor, and the acknowledgment that all cloud costs, overage charges, and account consequences are your own responsibility.

2.5 The Standard Chapter Template (One Chapter per Lab, LAB_01 to LAB_06)

Every lab chapter follows the same structure. The template file itself is re-distributed at Class 18; it is identical to what follows.

Chapter N — [Lab name] (LAB_0N)

N.1 Objective

Two or three sentences: what this lab defends, and which layer of defense-in-depth it adds to NorthMaple Accounting Inc.

N.2 Environment and prerequisites

Which machines, zones, and earlier chapters this procedure relies on.

N.3 Step-by-step procedure

Numbered steps (1, 2, 3, ...), exactly as actually performed.
One screenshot per key step. Every command written out in full.

N.4 Validation

The tests performed, the expected result, the observed result, and the evidence (screenshots, exports).

N.5 Problems met and fixes

Short troubleshooting notes: what went wrong, and what solved it.
("Nothing went wrong" is an acceptable entry.)

N.6 AI disclosure

The completed disclosure table of § 2.6, or the exact sentence:
"No AI tools were used in this chapter."

What each chapter must show (the deliverables named in the session plan):

Chapter	Must contain
1 — Defensible Network Architecture and Zoning	Environment setup; finished network diagram; zoning table with one-two sentence justifications
2 — Privilege and Access Management	Role and permission tables (Windows and Linux); validation checklist; test-login screenshots
3 — Host Hardening and Host Firewalls	Hardening checklist with before/after comparison; firewall rule tables; connectivity-test evidence
4 — IDS/IPS Deployment, Configuration, and Validation	Installation notes; the rule set used; alert screenshots from safe, contained test traffic
5 — Network Device Security and Network Firewall	Secured management access; firewall rule set enforcing the zoning; device registry; allowed/blocked validation evidence
6 — VPN Setup and Encryption Configuration	VPN configuration; encryption steps (data at rest); HTTPS with self-signed certificate; validation evidence

2.6 The AI Disclosure Format (Required in Every Chapter Where AI Was Used)

The academic-integrity rule of the Legal Notice (section 11): any use of an AI tool in a submission must state the LLM used and the **exact prompt(s)** given to it. Use this table in section N.6 of every chapter concerned:

#	AI tool (LLM) and version	Exact prompt given (verbatim)	Where the output was used	How the output was verified before use
1				
2				

Chapters where no AI tool was used state instead, word for word: **"No AI tools were used in this chapter."** A chapter that used AI without its disclosure is treated as an academic-integrity matter under the college's policies.

2.7 The Documentation Standard (Applies to Every Page)

1. **Plain, professional wording** — short sentences, no filler.
2. **Numbered steps** for every procedure.
3. **One screenshot per key step**, cropped to what matters, each with a one-line caption.
4. **Every abbreviation expanded in round brackets on first use** — for example: "DMZ (Demilitarized Zone)". This document follows the same rule; so must your manual.
5. **No real personal data and no real credentials, ever.** Screenshots show laboratory-only accounts and laboratory-only passwords; anything real (names of real people other than you, real e-mail addresses, real keys, cloud account identifiers) is redacted before submission (Legal Notice, section 14 — Privacy).
6. **Consistent naming** — machines, users, and files keep the same names across all chapters, matching the diagram of Chapter 1.
7. **Dates** — each chapter states when its procedure was performed.
8. **External links**, if any, are provided for convenience only and are verified before use (for example with the VirusTotal URL scanner, <https://www.virustotal.com/gui/home/url>); a link is not an endorsement (Legal Notice, section 10).

2.8 Closing Chapters — Resilience Summary and Lessons Learned

1. **Resilience assessment summary:** the completed ten-row template of § 1.5.7, plus a short paragraph stating, in plain language, how resilient the defended company is overall and which single improvement you would make next.
2. **Lessons learned:** one short paragraph — what you would do differently, and the most valuable thing the six labs taught you.

2.9 Grading Checklist — Course Procedure Manual (10 points = 10%)

Item	Points
Front matter complete: cover page, mandatory notice box, copyright line, company description, table of contents	1.0
Lab environment chapter: platform, VM list, isolation measures, cloud responsibilities where applicable	0.5
Chapter 1 (LAB_01) complete per template and reproducible	1.0
Chapter 2 (LAB_02) complete per template and reproducible	1.0
Chapter 3 (LAB_03) complete per template and reproducible	1.0
Chapter 4 (LAB_04) complete per template and reproducible	1.0

Item	Points
Chapter 5 (LAB_05) complete per template and reproducible	1.0
Chapter 6 (LAB_06) complete per template and reproducible	1.0
Resilience assessment summary and lessons learned	1.0
Documentation standard respected throughout (numbered steps, screenshots, abbreviation rule, no real data) and every AI disclosure present and complete	1.5
Total	10.0

Submission: one file (PDF or Word), inside the Final Project archive of § 1.6, named 420-D03-AB_C21_Manual_LASTNAME_Firstname.pdf (or .docx).

2.10 Why the Workload Stays Manageable

- The manual is written **during** the course, one chapter per lab, in the supervised work time already built into every class — not at the end.
 - The Final Project adds only assembly and one short checklist; all technical work is already done by Class 17.
 - Four full sessions (Classes 18 to 21) are reserved for project and manual completion, with a voluntary dry-run check at Class 20.
 - Combined with the ponderation of 3-3-3 (3 hours lecture, 3 hours practical work, 3 hours homework per week), a student who keeps up with the labs finishes both deliverables inside the scheduled time, without overload.
-

ANNEX A — Legal Notice, Disclaimer, and Acceptable Use Policy (Version 1.0) — Full Text

Author and owner	Cyber.SoHo
Version	1.0
Effective date	August 29, 2026
Applies to	Every lecture, lab, project, quiz, exam, document, video, and social-media publication of the course

A.1 Plain-Language Summary

This document protects everyone: the students, the viewers, and the author. In short: all the material of this course is provided for education only. You may practice the activities only inside your own, isolated laboratory environment, or on systems for which you hold explicit written permission. You are fully and solely responsible for your own actions, your equipment, your accounts, your costs, and your time. By reading, watching, downloading, or using any part of the material, you accept this entire document.

A.2 Educational Purpose Only

All lectures, labs, projects, quizzes, exams, documents, videos, and social-media publications of the course Cyber Defense (420-D03-AB) (together, "the Material") are created for one purpose: teaching defensive cybersecurity to adult learners of the Cybersecurity AEC (Attestation d'études collégiales) program. The Material shows how to protect systems, not how to attack the systems of others. Nothing in the Material is an invitation, an instruction, or an authorization to access, test, or damage any real system, network, account, or data.

A.3 Acceptable Use and Prohibited Actions (Student Warning)

You may:

- Practice only inside your personal, isolated laboratory environment, as defined in LAB_01: a local virtualization tool (for example VirtualBox) or your own personal free-tier cloud account on AWS (Amazon Web Services) or Microsoft Azure.
- Practice on any other system only if you hold explicit written permission from the owner of that system, obtained before you start.

- Use snapshots and isolated virtual networks so that a mistake stays inside your laboratory.

You must never:

- Scan, probe, attack, intercept, monitor, or access any system, network, wireless network, or account that is not yours — including those of the college, an employer, a business, a public website, or a neighbor.
- Create, install, or release malicious software outside your isolated laboratory.
- Attempt password, credential, or interception attacks against any real account or any real communication.
- Use the Material, or any tool presented in it, to break any law or any policy that applies to you.

Cloud responsibilities: if you use a cloud account, you must respect the acceptable-use policies of the provider (AWS or Microsoft Azure). Free tiers have limits; monitor them. All cloud costs, overage charges, and account consequences are your responsibility.

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Unauthorized computer access or use is a criminal offence. In Canada, the Criminal Code contains, among other provisions: section 342.1 (unauthorized use of a computer), section 342.2 (possession of a device to obtain unauthorized use of a computer), and section 430(1.1) (mischief in relation to computer data). Penalties can include imprisonment. If you are located outside Canada, the laws of your own country, province, or state apply to you, and you must know and obey them. The policies and codes of conduct of your college apply in addition to the law. The instructor never asks for, and never authorizes, any illegal act. If an activity seems to touch a system you do not own, stop and ask the instructor before doing anything.

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A.6 No Warranty

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A.7 No Professional Advice

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A.8 Intellectual Property

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A.11 Ethics and Responsible Conduct

The course teaches defensive ("white-hat") cybersecurity. By using the Material, students agree to use their skills only to protect, never to harm, and to respect the privacy of others. If a student accidentally discovers a real vulnerability, the student reports it responsibly to the owner of the system and to the instructor, instead of exploiting it or publishing it. Academic integrity applies at all times, including the course rule on AI (Artificial Intelligence): any use of an AI tool in a submission must state the LLM (Large Language Model) used and the exact prompt(s) given to it.

A.12 Social-Media Notice

Clips, posts, shorts, and videos published on public platforms (for example YouTube or TikTok) are educational demonstrations performed inside an isolated laboratory. Do not reproduce them outside such a laboratory. Third-party comments, re-uploads, edits, reactions, and translations are not verified and not endorsed by Cyber.SoHo, and they remain the sole responsibility of their authors. The terms of service and the community rules of each platform apply on that platform.

A.13 Indemnification

By using the Material, you agree to indemnify and hold harmless Cyber.SoHo and the instructor from any claim, demand, damage, loss, or expense (including reasonable legal fees) resulting from your own actions or omissions, from your violation of this document, or from your violation of any law or of any right of a third party.

A.14 Privacy

The Material itself collects no personal data. When an enrolled student signs the separate "Acknowledgment and Agreement" page, the student's name, the session number, the date, and a verification code are recorded in the course signature log; the drawn signature image stays only inside the record file that the student downloads and submits, and it is not published. If the signature log is hosted in a public repository, the log is publicly visible; hosting it in a private repository is recommended. On public platforms (for example YouTube or TikTok), the privacy policy of the platform applies.

A.15 Governing Law, Severability, and Updates

This document is governed by the laws applicable in the Province of Quebec, Canada. If a court finds one clause invalid, the other clauses remain in force. Cyber.SoHo may update this document at any time; the current version, identified by its version number, replaces all previous versions. Before this document is reused or republished outside the course, a review by a lawyer licensed in the Province of Quebec is recommended.

A.16 Acknowledgment and Acceptance

Reading, watching, downloading, or using any part of the Material means that you have read, understood, and accepted this entire document. Enrolled students also confirm their acceptance electronically: at the beginning of each session, they sign the separate "Acknowledgment and Agreement" page ([Acknowledgment_and_Agreement_eSign.html](#), provided separately on the course platform), which creates their personal electronic signature record for that session.

*Final Project and Course Procedure Manual — Full Content — Cyber Defense (420-D03-AB)
— Version 1.0 — Cyber.SoHo*

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